

Exploring issues of training data imbalance and mislabelling on random forest performance for large area land cover classification using the ensemble margin



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ARTICLE INFO

Article history:

Received 25 March 2014

Received in revised form 15 January 2015

Accepted 24 March 2015

Keywords:

Ensemble margin
Training data
Classification
Remote sensing
Imbalance
Mislabelling

ABSTRACT

Studies have demonstrated the robust performance of the ensemble machine learning classifier, random forests, for remote sensing land cover classification, particularly across complex landscapes. This study introduces new ensemble margin criteria to evaluate the performance of Random Forests (RF) in the context of large area land cover classification and examines the effect of different training data characteristics (imbalance and mislabelling) on classification accuracy and uncertainty. The study presents a new margin weighted confusion matrix, which used in combination with the traditional confusion matrix, provides confidence estimates associated with correctly and misclassified instances in the RF classification model. Landsat TM satellite imagery, topographic and climate ancillary data are used to build binary (forest/non-forest) and multiclass (forest canopy cover classes) classification models, trained using sample aerial photograph maps, across Victoria, Australia. Experiments were undertaken to reveal insights into the behaviour of RF over large and complex data, in which training data are not evenly distributed among classes (imbalance) and contain systematically mislabelled instances. Results of experiments reveal that while the error rate of the RF classifier is relatively insensitive to mislabelled training data (in the multiclass experiment, overall 78.3% Kappa with no mislabelled instances to 70.1% with 25% mislabelling in each class), the level of associated confidence falls at a faster rate than overall accuracy with increasing amounts of mislabelled training data. In general, balanced training data resulted in the lowest overall error rates for classification experiments (82.3% and 78.3% for the binary and multiclass experiments respectively). However, results of the study demonstrate that imbalance can be introduced to improve error rates of more difficult classes, without adversely affecting overall classification accuracy. © 2015 International Society for Photogrammetry and Remote Sensing, Inc. (ISPRS). Published by Elsevier B.V. All rights reserved.

1. Introduction

Accurate spatially explicit classification maps are important sources of information for natural resource land managers and forest monitoring programs. Land management agencies typically monitor and report on large areas (i.e. regional or continental scale, covering millions of hectares) relying on the interpretation of large complex remotely sensed data, calibrated and validated using, typically, a limited amount of ground reference data (Lippitt et al., 2008). Studies have demonstrated the successful application of ensemble machine learning classifiers, such as Random Forests

(RF), integrating remote sensing (satellite imagery) and ancillary spatial data, to improve supervised classification accuracy of forest and other natural environment land cover maps (Cutler et al., 2007; Mellor et al., 2013; Rodriguez-Galiano et al., 2012), for which conventional parametric statistical classification techniques might not be appropriate (Gislason et al., 2006). In ensemble classification, multiple (base) classifiers are constructed. From the ensemble, a final class is determined by, for example, averaging or a majority vote. In machine learning, the margin theory examines the proximity of data points to decision boundaries. Margin theory is a means by which to understand and evaluate ensemble classification and can be used to estimate confidence in the classification outcome (Schapire et al., 1998). Such ancillary information is important, particularly when relying on satellite image derived maps for scientific inference (McRoberts, 2011). The characteristics

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of training data is a fundamental consideration when constructing any supervised classifier (including ensemble machine learning). Learning from imbalanced training data (i.e. unevenly distributed data between classes) is a common problem (Japkowicz and Stephen, 2002). Machine learning algorithms, such as RF, are constructed to minimize the overall classification error rate and imbalanced training data can result in poor accuracy for minority classes (Chen et al., 2004). Furthermore, it is assumed that, in its implementation, the classifier is run using data drawn from the same distribution as the training data (Provost, 2000). In RF, decision trees are induced using bootstrap samples of training data (Breiman, 2001) and in situations where training data includes only a minority of training data samples for a particular class (relative to other classes), it is likely that a bootstrap sample may include few or even no samples from this class and hence fewer leaves describing the minority class, resulting in poor classification accuracy for the minority class prediction (Chen et al., 2004) as well as weaker confidence estimates (He and Garcia, 2009).

The imbalance training data problem is common in large area natural resource applications using remote sensing (e.g. forest classification), whereby within reference data, rare land cover or forest classes may be under-represented relative to more abundant classes, due to the time and cost resource constraints of collecting enough representative training samples. Studies have shown balanced datasets improve overall classification compared to imbalanced data (Estabrooks et al., 2004; Weiss and Provost, 2003). Several techniques have been demonstrated to address the imbalance training data problem. These include down-sampling majority classes (Freeman et al., 2012) and weighting rare training observations more highly than common classes (Chen et al., 2004). Techniques involving over-sampling the minority class through replication of samples to match the quantity of majority class training samples (Ling and Li, 1998) and a combination of over-sampling (minority) and down-sampling (majority) training classes (Chawla et al., 2002) have also been explored.

Training data class mislabelling (or noise) is another important consideration in using bagging ensemble algorithms such as RF. This is an issue that often adversely affects machine learning algorithms (Guo, 2011). In large area remote sensing classification for forest monitoring programs, training data typically include ground-based (i.e. field data collection) (Lillesand and Kiefer, 1994) or data sampled from remote sensing imagery of a higher spatial resolution, such as very high resolution satellite imagery (e.g. Quickbird) or digital aerial photography (Wulder, 1998). Deriving training data using manual and semi-automated mapping from high spatial resolution imagery are methods which are prone to a variety of sources of labelling error and bias. These sources include interpreter bias and inconsistency, spatial resolution (scale), geometric and radiometric variability, and error associated with temporal discontinuity between training data (i.e. aerial photography acquisition date or season) and satellite imagery used for classification (Morgan et al., 2010). Other training data labelling errors are associated with inconsistency of vegetation classification methods, techniques and spatial resolution (Bradley and Friedl, 1996). In forest environments, common training data class mislabelling errors are caused by the similarity of forest types as their signatures appear in aerial photography (Delaney and Skidmore, 1998).

For their application in an operational setting (such as a large area forest monitoring program), it is important that machine learning classifiers are resilient to mislabelling in training data (Lippitt et al., 2008). Studies have demonstrated the relative resilience of bagging ensemble classifiers, such as RF, to training data noise (class mislabelling) (DeFries and Cheung-Wai Chan, 2000). In evaluating machine learning algorithms for land cover change mapping, Rogan et al. (2008) investigated the effect of artificially

introduced training data noise to classification accuracy. Their study found the addition of 10% noise reduced accuracy of decision tree classifiers S-Plus and C4.5 by 7% and 20% respectively. In a land cover classification study, Rodriguez-Galiano et al. (2012) found the RF classifier performance (overall classification error) to be relatively insensitive for up to 20% deliberately mislabelled training instances, above which error rate increased exponentially. Na et al. (2009) reported a reduction in RF overall accuracy by almost 50% associated with a 30% increase in the amount of artificial noise.

In this paper, we examine how training data class imbalance and class mislabelling affect RF performance in the context of large area forest classification in an operational land management agency setting. This was achieved across diverse and complex forest ecosystems and topography, dominated by open canopy sclerophyll forests and woodland. We evaluate RF performance associated with training data characteristics through a new perspective involving ensemble margins. The magnitude of ensemble margin is usually interpreted as a measure of confidence in classification prediction and significant work has been published about bounding and reducing prediction error based on the classification margin (Guo, 2011; Schapire et al., 1998). The nature of a training set can have a major impact on classification accuracy (Foody, 1999) and the margin ensemble can be used to understand how training data characteristics can affect classification outcomes. Foody (2002) emphasizes the need for more accuracy assessment information (including confidence measures) to be provided with land cover and other remote sensing derived classification maps, to aid user interpretation and application. The value of very large area mapping is ultimately limited by poor quality accuracy assessment and reporting (Foody, 2002). In this study, we evaluate new ensemble margin statistics as a means of providing distinct information about margin distribution and classification prediction confidence and supplementing traditional measures of classification performance. Furthermore, we introduce a novel method for assessing classification uncertainty through the use of an ensemble margin weighted confusion matrix, that to the best of our knowledge is used for the first time in land cover classification using remote sensing and ancillary geospatial data.

2. Random Forests

Random Forests (RF) uses a bootstrap aggregation technique (bagging) (Breiman, 1996) to generate sub-sets of training data with which to build an ensemble of decision trees (base classifiers). The bagging process involves resampling the original training set with replacement, resulting in a greater diversity of decision trees, thereby improving classifier stability and accuracy. Moreover, in constructing trees, as some training data instances may be used more than once or not at all, correlation between trees is reduced, and as a result, RF is more robust to variations in input data and less sensitive to mislabeled training data or over-fitting (Pal, 2005; Rodriguez-Galiano et al., 2012).

In constructing each decision tree, at each node (split) a randomly selected subset of model predictor variables are evaluated for partitioning the data into increasingly homogeneous subsets – the variable used to split the data is that which results in the greatest increase in data purity. Increasing the number of predictor variables selected for tree construction results in stronger individual decision trees, but with increased correlation between trees, model accuracy is reduced (Rodriguez-Galiano et al., 2012). Therefore, to minimize the generalization error, it is necessary to optimize this parameter, together with the number of decision trees in the ensemble. Tree building continues until there are no further gains in purity. A response variable can be predicted as an average (continuous variable classification) or model vote

(categorical classification) among all decision trees. Sample Out of Bag (OOB) data (that are not drawn in the bagging sample used for tree construction) are used to compute accuracies and error rates averaged over all predictions (Cutler et al., 2007).

3. Ensemble margin

The margin function is an important concept in ensemble classifiers such as RF. The classical margin function (Schapire et al., 1998) is calculated for each known data instance, and ranging from -1 to $+1$, is the normalised difference between the votes assigned to the true class and those assigned to the most voted class that is different from the true class (Guo et al., 2011). An alternative margin function, which does not require the known class labels, is an unsupervised version of Schapire's margin. It combines the first and second most probable class labels under the model. Eq. (1) shows how the unsupervised margin is calculated where V_{c1} represents the number of votes for the most voted class c_1 for instance x and V_{c2} the number of votes for the second most popular class c_2 . The unsupervised margin ranges from 0 to 1. Instances close to class boundaries (margin values around 0, i.e. not redundant and which are located near decision boundaries) are the most informative for the classification task. In that case, the true class labels are not of significance. As such, the unsupervised margin may be more robust to class mislabelling (Guo, 2011). Hereafter, unless otherwise specified, the term margin is used to describe the unsupervised margin.

$$\text{margin}(x) = \frac{V_{c1} - V_{c2}}{\sum_{c=1}^L (V_c)} \quad (1)$$

This may be also be expressed as:

$$\frac{\max_{c=1, \dots, L} (V_c) - \max_{c=1, \dots, L \cap c \neq c_1} (V_c)}{T}$$

where T represents the size of the ensemble and L represents the number of classes

The margin of a correctly classified instance should be as high as possible (close to 1) and the margin of a misclassified instance, as low as possible (close to zero). Indeed, the lower the margin of a misclassified instance, the greater the opportunity for improving the classification by the ensemble of base classifiers.

4. Study site and data

4.1. Study area

The study area (Fig. 1) comprises approximately 7.1 million hectares of public land covering about one third of the state of Victoria, south east Australia. The extents of Victoria are north $141^\circ 47' 36''$ E $33^\circ 58' 54''$ S, east $149^\circ 58' 36''$ E $37^\circ 30' 20''$ S, south $146^\circ 17' 13''$ E $39^\circ 9' 33''$ S and west $140^\circ 57' 29''$ E $34^\circ 28' 23''$ S. The study area includes two public land tenures – national parks and conservation reserves and multiple-use commercial State forests, and covers a wide range of ecosystem types and a high diversity of forest types and structures, dominated by open sclerophyll forests. The state is characterised by diverse topography and a range of climate zones.

4.2. Reference (training and test) data

On-screen digital Aerial Photographic Interpretation (API) of 30–50 cm ground sample distance colour aerial photographs (photoplots), acquired between 2006 and 2010, was the source of classification reference data (used for model training and test validation data). Seven hundred and sixty-six 2×2 km photo-plots,

systematically distributed across a state-wide random stratified grid (Fig. 1), were used to produce land cover maps based on a land cover classification system (Mellor and Haywood, 2010) which included broad forest or land cover type, forest canopy height class (low, medium, tall) and canopy cover class (woodland, open, closed). Thresholds for forest canopy height and cover are shown in Fig. 2. Land cover object delineation was undertaken by trained interpreters, using photo-plot information including crown shape, size and arrangement, colour and shadow. Forest delineation followed the Australian National Forest Inventory forest definition (greater than 20% crown cover, minimum two metre stand height) (Department of Agriculture Fisheries and Forestry, 2012) with a 0.5 ha minimum mapping unit applied to the land cover maps, based on the UNFAO forest definition (Food and Agriculture Organization of the United Nations, 2001). Ancillary GIS polygons representing the boundaries of clear-fell logging and wild fires were used to update land cover polygons to a summer season 2008/09 baseline. A comprehensive description of the API land cover mapping method can be found in Farmer et al. (2013).

For this study, land cover data were aggregated into three forest canopy cover classes (woodland, open, closed), shrub (woody vegetation not meeting the 2 m height threshold or 20% crown cover) and other non-forest land cover, and binary forest/non-forest classes. Land cover polygons were converted into raster format aligned with the resolution of classifier predictor variables described in the following section. Test (validation) data were randomly selected, using simple random sampling, from the total reference data and set-aside prior to selection of training data for building RF models.

4.3. Predictor variables

Predictor data comprised multi-source remote sensing, topographic and climate variables. Landsat TM data are commonly used in studies for forest type discrimination and cover estimation (Boyd and Danson, 2005). In this study, a 6-band study-area Landsat TM mosaic comprising nineteen scenes acquired between February and March 2009, was used. The time of acquisition was designed to correspond with the training and test data (land cover mapping) and late summer conditions, where high sun angles reduce shadow and terrain effects and where spectral reflectance differences between overstory evergreen woody vegetation and seasonally dynamic understory vegetation is maximized (Mellor et al., 2013). Following Flood et al. (2013), Landsat TM scenes were standardized to surface reflectance to reduce sources of inter-scene variation associated with atmospheric conditions, topography, sensor location and sun elevation.

Where vegetation appears spectrally similar but has differing spatial patterns, textural indices derived from satellite imagery have been shown to improve classification performance (Kayitakire et al., 2006; Rodríguez-Galiano et al., 2011). For this study, a grey-scaled (8-bit) Normalized Difference Vegetation Index (NDVI) layer was generated using Landsat TM bands 3 and 4, from which were produced textural indices representing spatial variation in optical imagery (Haralick, 1979). First order texture measures of variance and diversity were generated for 3×3 and 5×5 cell neighbourhood moving windows, designed to capture textural variance of the study area's forested ecosystems (Mellor et al., 2013).

Multi-temporal MODIS NDVI data was used to represent phenological variance in the study area over a calendar year. Twenty-three 250 m spatial resolution 16-day MODIS NDVI images from 2008, covering the study area, were extracted from an Australian national mosaic (Paget and King, 2008), from which one standard deviation raster was calculated using each multi-temporal image pixel stack – to represent seasonal variance in NDVI.

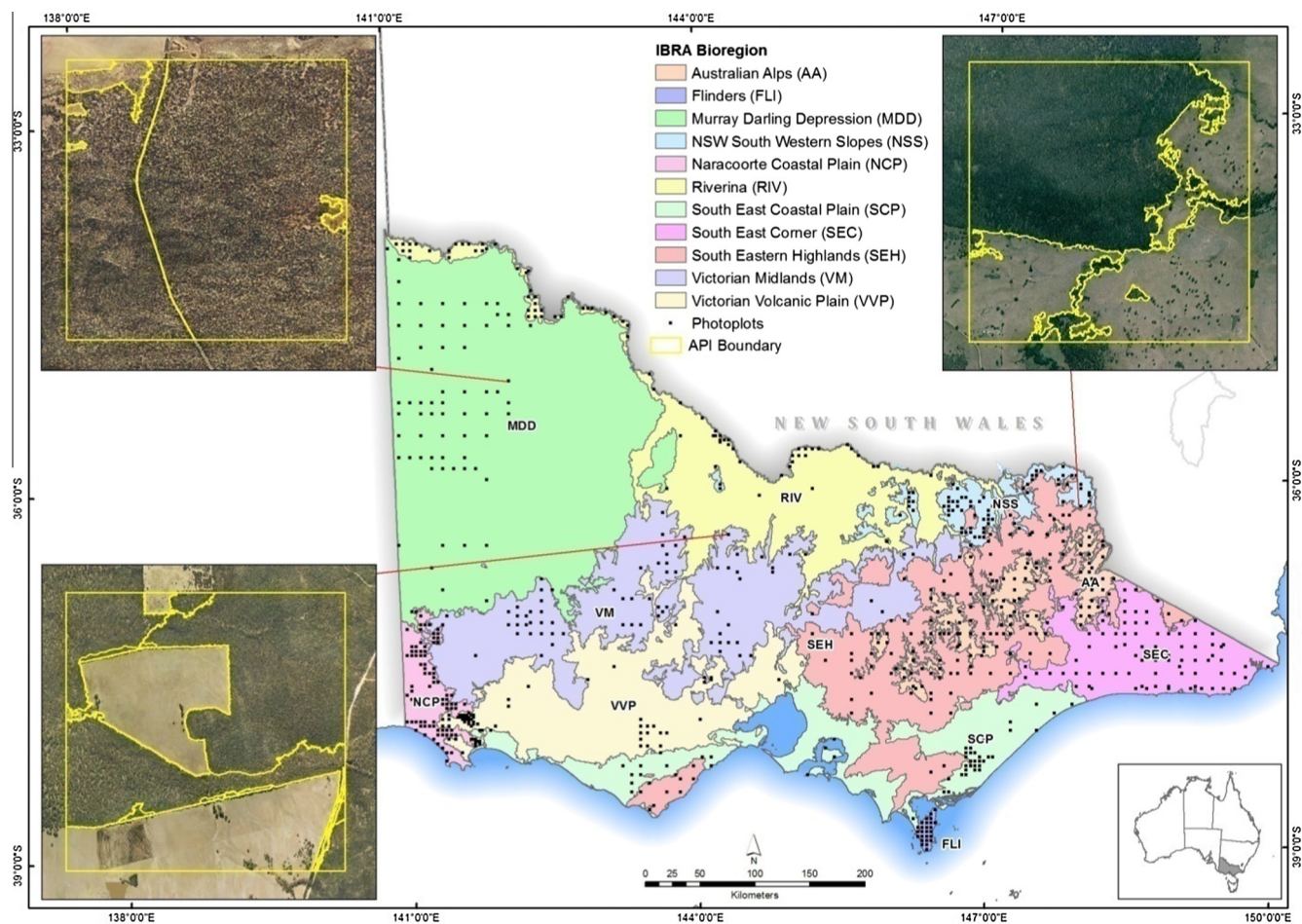
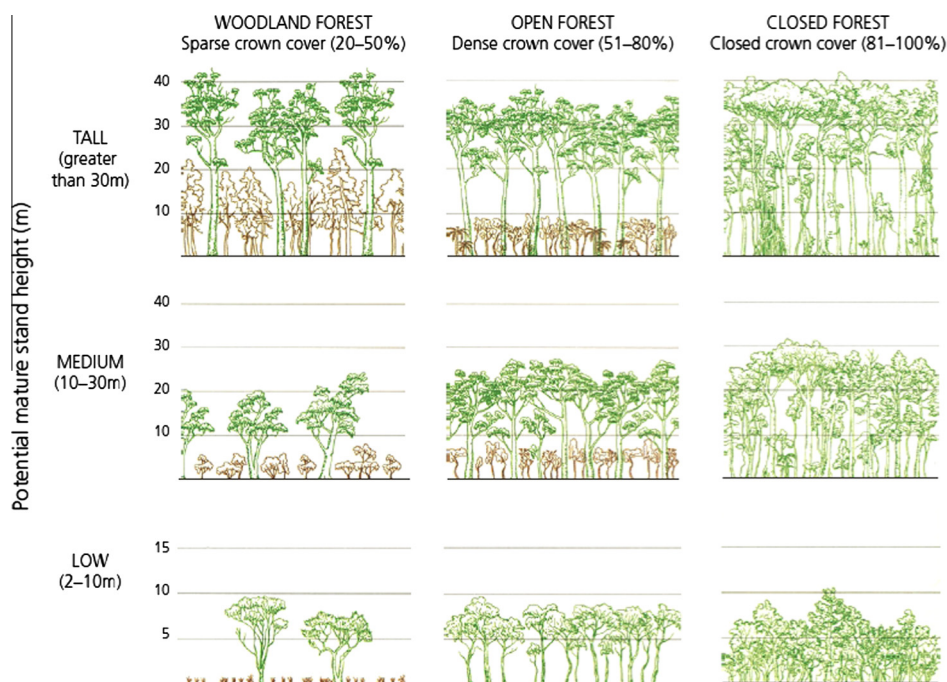


Fig. 1. Victorian Interim Biogeographic Regionalisation for Australia (IBRA Bioregions) and Aerial Photographic Interpretation (API) land cover maps.



Source: Australian Land Information Group and JA Carnahan (1990). *Atlas of Australian Resources, Vegetation*. Australian Government Publishing Service, Canberra.

Fig. 2. Australian forest structural definitions used in the study.

Topographic and biophysical information was included in the model as a means of capturing species–environment relationships, which are central to predictive geographical modeling (Guisan and Zimmermann, 2000; Mellor et al., 2013). Bioclimatic maps support assumptions about the influence climate has on forest type distribution (Beaumont et al., 2005) and the composition of vegetation that can be expected to occur in areas with similar soils, climate and topography (Franklin, 1995). Following Gislason et al. (2006), topographic predictor variables of elevation (metres), slope and aspect (degrees) were included in the model, derived from a 1 s (~30 m) smoothed Digital Elevation Model (DEM) (CSIRO, 2011). The DEM was used to generate precipitation, temperature, radiation and moisture climate prediction surfaces using BIOCLIM in the ANUCLIM (v 5.1) software package (Houlder, 2001). A detailed description of the BIOCLIM modelling process can be found in Beaumont et al. (2005).

Reference data vector polygons were converted into raster format and resampled, using the nearest neighbour method, together with the predictor variables, to align with the resolution and extent of the 30 m × 30 m Landsat TM raster cells.

5. Methods

5.1. Experiments

Five experiments were carried out, using binary (experiments 1 and 4) and multiclass (experiments 2, 3 and 5) models, to examine the effect of training data class imbalance and mislabelling on RF performance. Binary classification experiments used forest/non-forest categories. Multiclass experiments used canopy cover classes (woodland, open and closed), together with two non-forest classes (shrub and other non-forest). For each experiment, two training data sizes (subsets) were established: optimal and critical (Table 1). The optimal subsets for binary and multiclass experiments were estimated by running multiple iterations of the RF classifier with a balanced distribution of training samples per class, increasing the training set size with each iteration. Classification accuracy typically increases with training data size until reaching an asymptotic level (constant accuracy) indicating that the optimal accuracy has been achieved. The optimal training size is the minimum training size leading to this maximum accuracy, hence it is optimal in terms of both classification accuracy and complexity. Critical subset sizes for binary and multiclass experiments were 1% and 5% of the optimal subsets respectively. To ensure stable accuracy estimates from critical training sets, ten iterations of each experiment were run, from which mean performance measures were calculated.

5.1.1. Training data class imbalance experiments

An examination of the impact of class imbalance on RF model performance was undertaken on training sets for binary (experiment 1) and multiclass (experiments 2 and 3) classification. For the binary imbalance experiment (1), for each RF model, balance as a ratio of forest to non-forest training samples was adjusted while maintaining the same total number of training samples.

For the multiclass (forest cover) classification experiments (2 and 3), an initial RF model was generated using a class-balanced distribution of training data. Canopy cover classes from the forest super class (comprising woodland, open and closed forest classes) with the highest and lowest producer's accuracies (omission errors) were identified from a confusion matrix generated with test data. Hereafter these are referred to as *best* class (i.e. easiest class) and *worst* class (i.e. hardest class).

The first multiclass imbalance experiment (2) adjusted the ratio of *best* to *worst* class training samples in each RF model. The second multiclass imbalance experiment (3) involved generating imbalance in the training data samples by increasing the proportion of the *worst* class while simultaneously decreasing the proportion of the *best* class by the same amount. This is a sensible strategy that can outperform a balanced distribution at least in terms of per class accuracies as will be shown later by our experiment results. Hence, balancing training data is not always the best strategy especially if the classification task involves classes of differing complexity. For every iteration of both multiclass imbalance experiments, the number of samples representing the remaining classes was kept constant, thereby maintaining the same total number of multiclass training samples for each experiment.

For the binary classification, imbalance experiment (1) involved adjusting balance as a ratio of forest to non-forest training samples (10:90, 25:75, 50:50 (balanced), 75:25 and 90:10) while maintaining the same total number of samples (10,000 and 100 for optimal and critical sizes respectively). For the multiclass classification experiments (2 & 3), an initial RF model using a class-balanced training data distribution, determined the closed canopy cover and open canopy cover classes to be the *best* (easiest) and *worst* (hardest) of the forest cover super class. These classes were adjusted in the multiclass imbalance experiments. Firstly by changing the ratio of open to closed cover class samples (10:90, 25:75, 50:50 (balanced), 75:25 and 90:10), while maintaining the same total number of samples including each of the remaining classes (woodland, shrub and other non-forest) for both optimal (100,000 samples) and critical (5000 samples) cases. In the second multiclass imbalance experiment, the proportion of open and closed cover classes in each model was adjusted by increasing the proportion of the Open (worst, or most difficult) class (5%, 10%, 25%, 50%, 75% and 90%), while simultaneously decreasing, by the same proportion, the number of Closed (best, or easiest) class training samples (Table 2).

5.1.2. Training data class mislabelling experiments

For the binary classification mislabelling experiment (4), training data mislabelling was undertaken by randomly re-assigning a proportion of forest instances as non-forest and non-forest instances as forest. While introducing artificial noise into binary class training data is a straightforward process, it is not always the case for multiclass problems. Studies typically use random class-label switching to simulate noise in a classification (Guo, 2011; Rodriguez-Galiano et al., 2012; Rogan et al., 2008; Sluban et al., 2014). We propose an alternative approach, designed to replicate realistic real-world operator misclassification (mislabelling) of reference data instances (Lowell et al., 2005), that results in a more reliable analysis of noise effects on a supervised classifier performance. In the multiclass mislabelling experiment (4), a preliminary RF model was built using a balanced distribution of samples of the multiclass training data. A confusion matrix derived from the OOB data (not used in the bootstrap training samples) was used to determine, for each class c_i , the class to which it was most frequently misclassified l_i , i.e. the most frequent error class predicted by the model from the OOB data. For the multiclass classification, starting with a training data set with more or less

Table 1
Optimal and critical training and test set sizes used for binary and multiclass experiments.

Classification	Training data (total samples)		Test data
	Critical	Optimal	
Binary (experiments 1 & 4)	100	10,000	2500
Multiclass (experiments 2, 3 & 5)	5000	100,000	25,000

Table 2

Training set sizes for each class for multiclass imbalance (experiments 2 and 3).

Class	Training set size per class	% Increase in <i>worst</i> class and decrease in <i>best</i> class					
		5	10	25	50	75	90
<i>Optimal</i>							
Woodland	20,000	20,000					
Open (worst)	20,000	21,000	22,000	25,000	30,000	35,000	38,000
Closed (best)	20,000	19,000	18,000	15,000	10,000	5000	2000
Shrub	20,000	20,000					
Non-forest	20,000	20,000					
<i>Critical</i>							
Woodland	1000	1000					
Open (worst)	1000	1050	1100	1250	1500	1750	1900
Closed (best)	1000	950	900	750	500	250	100
Shrub	1000	1000					
Non-forest	1000	1000					

“real” noisy labels whose amount is unknown in practice, the introduction of artificial noise in class labels is performed by mislabelling a proportion of each class c_i to l_i ($c_i \rightarrow l_i$). For this experiment, iterations of the multiclass model were run, mislabelling an increasing proportion of each class c_i to l_i each time. Based on the results (confusion matrix) of a *clean* balanced training set, these were as follows:

- Woodland $\rightarrow$ Open
- Open $\rightarrow$ Closed
- Closed $\rightarrow$ Open
- Shrub $\rightarrow$ Woodland
- Non-forest $\rightarrow$ Woodland

5.2. Random forest model parameters

The *randomForest* package (Liaw and Wiener, 2002) in R (R Development Core Team, 2011) was used to build the RF models. For all models, the default number of randomly sampled variables as candidates for each decision tree node split was used (equal to the square root of the total number of predictor variables). For binary classification experiments, RF decision tree ensembles were constructed with 100 trees, a typical moderate size ensemble (Tsoumakas et al., 2009). For the more complex multiclass case, more trees were used in each ensemble (150). Assignment of class was determined by the majority of votes from all decision trees in the ensemble, a standard approach for combining the decisions of multiple component learners.

5.3. Random forest model performance evaluation

2500 Samples (1250 per class) were used as test data for binary classification experiments, and 25,000 (5000 per class) for multiclass experiments (Table 2). Test data were used to calculate overall and per-class accuracies and Kappa coefficient. Kappa is a measure of accuracy adjusted for chance agreement (Carletta, 1996). Kappa coefficient ranges from 0 to 1.0, with 1.0 representing 100% better agreement than by chance alone.

We introduce three ensemble margin descriptive statistics (mean, dominant (frequency) mode, entropy) – measures to analyse the effect of imbalance and mislabelling on RF. Classification uncertainty is assessed through the use of a novel weighted confusion matrix based on ensemble margin. These measures were calculated from unsupervised margin values (equation 1) of each model. Cumulative frequency distribution curves of correct and misclassified instance margins were also used to illustrate and compare confidence rates between different models and experiments.

5.3.1. Mean margin

A margin criterion measuring a strong ensemble classifier is one which maximises the mean margin of correctly classified samples and minimises the mean margin of misclassified samples. This normalised measure, ranging from -1 (weakest classifier) to $+1$ (strongest classifier), is defined as follows Eq. (2):

$$\mu = \frac{(n_c \mu_c) - (n_m \mu_m)}{n_c + n_m} \quad (2)$$

where n_c is the number of correctly classified instances, n_m is the number of misclassified instances and μ_c and μ_m are mean margins for correctly and misclassified instances respectively.

5.3.2. Dominant margin

The dominant margin is based on the mode of the margin histogram (margin value with highest frequency). The margin mode is calculated as follows:

1. Group margin values into bins (10 bins were used in this study): $[0,0.1]$, $[0.1,0.2]$, ..., $[0.9,1]$.
2. Calculate margin bins histogram.
3. Detect the peak of the histogram (the bin with the highest frequency).

A normalised measure of the dominant margin is calculated from two margin modes, one for the margin distribution of correctly classified instances and another for misclassified instances Eq. (3).

$$M = \frac{n_c M_c - n_m M_m}{n_c + n_m} \quad (3)$$

where M_c and M_m are the margin modes of correctly classified and misclassified instances respectively.

5.3.3. Margin entropy

Shannon's entropy (Shannon, 1948) was used to measure diversity and redundancy in margin distribution, applied on margin normalised frequency values. The margin entropy is estimated using the following steps:

1. Group margin values into bins.
2. Calculate margin bins histogram.
3. Normalise the histogram to determine bin (or coarse margin) probabilities for each margin bin.
4. Calculate entropy on resulting bin probabilities using Shannon's formula Eq. (4)

$$H = -\sum_{i=0}^{n-1} [P(m_i) * \log_2(P(m_i))] \quad P(m_i) \neq 0 \quad (4)$$

where P represents probability, m_i ($0 \leq i < n$) a margin bin and n the number of margin bins.

The minimum value for margin entropy is 0 (lowest diversity). The entropy is maximum when underlying events are equiprobable, i.e. all margin frequencies are the same. For 10 margin bins, the maximum value of margin entropy is 3.32 (highest diversity such as random distribution). Unlike mean and dominant margins, the introduced margin entropy, which expresses diversity in ensemble models at data level, has to be high, but not at its maximum. Indeed, more diversity does not necessarily induce higher classification accuracy (Kuncheva and Whitaker, 2003). The complex relationship between diversity and ensemble accuracy is still not well understood, but diversity is recognised as an indispensable condition in designing effective ensemble classifiers (Kuncheva and Whitaker, 2003).

5.3.4. Margin-weighted confusion matrix

We introduce a novel Weighted Confusion Matrix (WCM) that uses the unsupervised margin of ensemble classifiers. It has been recommended that traditional accuracy estimates should be accompanied by confidence limits, which are rarely provided in published papers (Foody, 2004). This WCM, used together with a confusion matrix, provides a more thorough analysis of multiple classifier performance than a traditional deterministic approach based solely on a confusion matrix. The Margin-Weighted Confusion Matrix (MWCM) measures the degree of certainty associated with correctly and misclassified instances in the RF model. The MWCM is calculated as follows:

1. Calculate the unsupervised margin of each instance Eq. (1).
2. Assign to each instance its reference and RF model predicted class labels.
3. For every reference-to-predicted class combination (e.g. woodland predicted as woodland, woodland predicted as open, etc.) calculate the sum of margins and the count of instances.
4. Populate each confusion matrix cell with the sum of margins divided by the count of instances in the cell (normalisation).

The ideal MWCM is an identity matrix (i.e. $L \times L$ square matrix with ones on the main diagonal and zeros elsewhere), and the closer a model MWCM to this configuration, the stronger the underlying ensemble classifier. The classification confidence of correctly classified instances (MWCM main diagonal cells) should be as high as possible (close to 1). Conversely, the confidence of misclassified instances (MWCM non-diagonal cells) should be as low as possible (close to 0).

6. Results and discussion

6.1. Effect of training data imbalance on RF performance

6.1.1. Experiment 1: Binary imbalance

The effect of training data class imbalance on binary classification performance is shown in Tables 3 and 4, and in Fig. 3. Overall accuracy was highest in the balanced case for both optimal (overall accuracy 91.16%, Kappa 82.32%) and critical (overall accuracy 84.61%, Kappa 69.22%) cases. Results demonstrate that imbalance increases per class accuracy in favour of the majority class, and at the expense of the minority class (Table 3), as fewer training instances of the minority class are selected in each bootstrap sample used in tree construction. This effect is more pronounced in the critical case, where very few training samples between highly

Table 3

RF model performance results for binary classification imbalance (experiment 1).

Balance, ratio Forest to Non-Forest training samples					
	10:90	25:75	50:50	75:25	90:10
Overall Kappa (%)					
Optimal	60.8	78.24	82.32	77.44	66.4
Critical	30.74	62.34	69.22	61.02	37.26
Margin entropy					
Optimal	2.85	2.78	2.61	2.54	2.52
Critical	2.97	3.09	3.02	2.93	2.70
Optimal					
Per-class producer Kappa (%)					
Forest	45.02	70.16	84.28	92.19	95.18
Non-forest	93.6	88.43	80.45	66.76	50.98
Critical					
Per-class producer Kappa (%)					
Forest	18.68	52.46	77.32	87.66	93.12
Non-forest	86.89	76.82	62.66	46.8	23.29

Table 4

Binary imbalance confusion matrices and margin-weighted confusion matrices for evenly balanced and imbalanced training data in optimal and critical cases.

Forest to Non-forest training sample balance	Observed (test) data			
	Confusion matrix		Margin-weighted confusion matrix	
	Forest	Non-forest	Forest	Non-forest
Optimal				
Prediction				
50:50 (Balanced)				
Forest	1149	127	0.80	0.49
Non-forest	101	1123	0.37	0.81
25:75				
Forest	1054	70	0.66	0.42
Non-forest	196	1180	0.49	0.88
Critical				
Prediction				
50:50 (Balanced)				
Forest	1095	237	0.70	0.47
Non-forest	155	1013	0.39	0.71
25:75				
Forest	842	102	0.53	0.37
Non-forest	408	1148	0.47	0.82

imbalanced minority cases (e.g. 90% forest to 10% non-forest) results in large differences in majority and minority class accuracies (59.9% difference in the critical case versus 30.2% difference in the optimal case). Nevertheless, the balanced case provides the best (more balanced) pairwise per-class accuracies for both optimal and critical training set sizes. The minimum accuracy per class for an optimal training set size is 90% (80.45% Kappa) versus 83.36% (70.16% Kappa) for the best imbalanced case performance (25:75).

Mean margins, measuring the overall classification confidence, are also highest where training data are balanced (Fig. 3b). Confusion Matrices (CM) and Margin Weighted Confusion Matrices (MWCM), comparing optimal and critical balanced and imbalanced (25:75) experiments, are shown in Table 4. The outcome of both class label prediction and underlying uncertainties has significantly evolved in favour of the majority class (non-forest). The non-forest accuracy increased by 4.88% while the forest accuracy decreased by 8.9% (almost a factor of 2). The MWCM shows an increase in non-forest classification confidence of 7% and a decrease in the certainty of misclassifying non-forest as forest by the same amount. However, the class imbalance led to poor classification confidence for the minority class: loss in

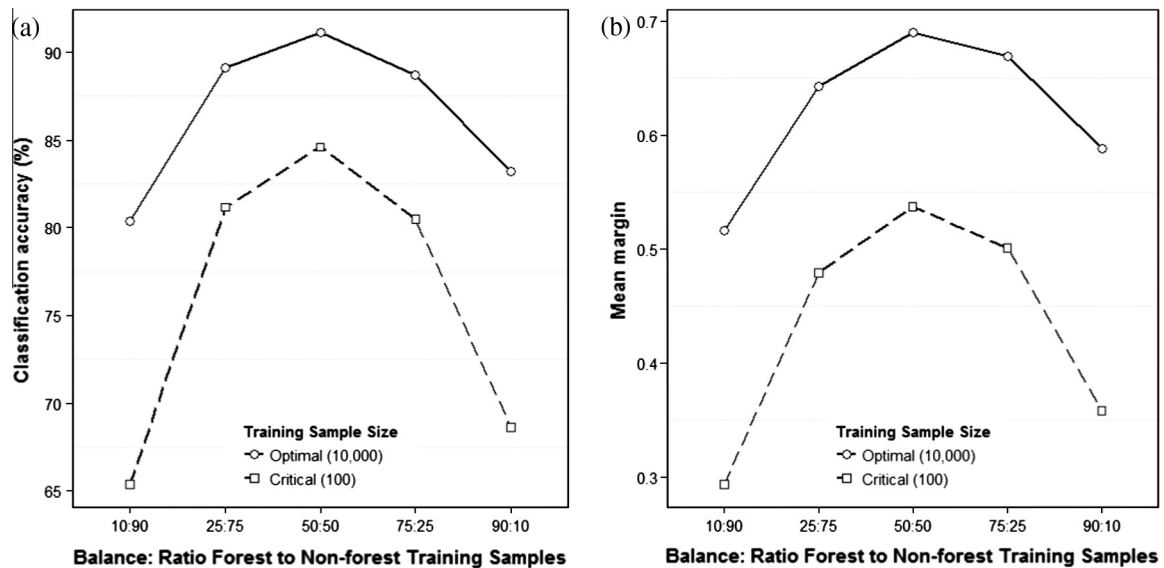


Fig. 3. Effect of binary class imbalance on overall classification accuracy (a) and mean margin (b).

classification confidence of 14% and increase in uncertainty associated with forest to non-forest confusion of 12%. Hence, the RF performance loss for the minority class (forest) is twice as big as the gain for the majority class (non-forest) in both classification accuracy and associated confidence. These class imbalance effects are again, more pronounced in the critical case (Table 4).

Unsupervised margin cumulative distribution curves (Fig. 4) for balanced training data, illustrate the higher degree of uncertainty associated with the correctly classified instances for critical experiment and the lower certainty associated with misclassified instances for the optimal case. Furthermore, the critical margin distribution of correctly classified instances exhibits a significantly higher proportion of low margins (<0.5). Meanwhile, the critical margin distribution of misclassified instances exhibits a lower

proportion of low margins. Both reflect poorer behaviour of the underlying RF model.

The margin entropy values Eq. (4) are higher for critical than for optimal size (e.g. 3.02 vs. 2.61 for the balanced case) (Table 3). This is expected as less data increases uncertainty leading to greater diversity. The interpretation of this concept has to be carefully addressed. For instance, for the critical case, the margin entropy value for ratio 50:50 is already high: 3.02 (maximum at around 3.32 for 10 margin bins). It increases to 3.09 for ratio 25:75. In the meantime, the mean margin decreases (Fig. 3b), reflecting an increase in lower margin frequencies. The increase in margin entropy suggests an initially higher proportion of higher margins at ratio 50:50 which becomes more balanced at 25:75. From ratio 25:75 to 10:90, the entropy decreases while the mean margin continues to decrease. This means that the behaviour is reversed: the proportion of lower margins has become dominant with respect to higher margins. Among the five tested ratios, 25:75 results in maximum entropy (and hence maximum diversity) but it is the balanced case (50:50) which results in maximum accuracy. This is consistent with entropy theory outlined in Section 5.3.3.

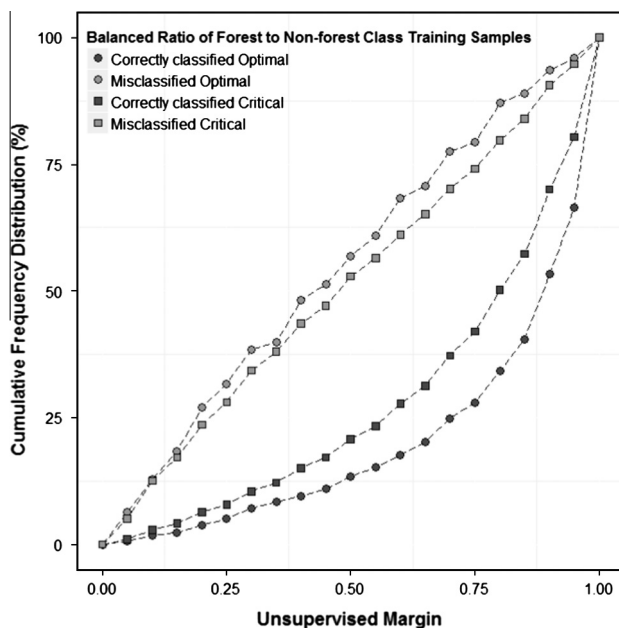


Fig. 4. Binary classification unsupervised margin cumulative frequency distribution curve, comparing correctly and misclassified instance confidence, for optimal and critical training sizes.

6.1.2. Multiclass imbalance (experiments 2 and 3)

For the multiclass model (experiment 2), in the balanced optimal case, per class (producer) accuracies were highest for the closed canopy cover (89%), shrub (88.9%) and Non-forest (87.8%) classes. Increasing the proportion of the most difficult class (open canopy cover) whilst simultaneously decreasing the proportion of the easiest class (closed canopy cover), improved class accuracy and Kappa for the most difficult (open cover) class (Table 5). Compared to the balanced class case, a 25% increase in open class samples and decrease in closed class samples resulted in 7.5% gain in producer accuracy for the open class, but a 9% reduction in producer accuracy for the closed class, leading to more balanced pairwise (Open/Closed) per-class accuracies. However, despite this introduced imbalance, overall accuracy was only marginally affected (1.2% less than the balanced model). Similarly, this imbalance resulted in only a minor reduction in per class accuracy for the shrub and non-forest classes (1.4% and 0.3% respectively for a 25% imbalance). A 10% increase in open class samples and decrease in closed class samples led to the best performance (overall producer accuracy). Indeed, this resulted in an increase in the minimum accuracy per class by 1.22% (1.56% Kappa) compared to the

Table 5

RF model performance results for optimal and critical multiclass classification imbalance experiments.

% Increase worst class (open) and Decrease best class (closed)							
	Balanced	5	10	25	50	75	90
<i>Overall Kappa (%)</i>							
Optimal	78.3	77.64	77.25	76.78	73.96	68.26	62.23
Critical	62.61	61.85	61.84	60.91	57.81	51.77	48.04
<i>Margin entropy</i>							
Optimal	3.24	3.24	3.23	3.24	3.24	3.25	3.25
Critical	3.19	3.19	3.17	3.17	3.17	3.24	3.30
<i>Optimal</i>							
<i>Per-class producer Kappa (%)</i>							
Woodland	69.04	68.52	67.6	65.89	63.8	60.93	59.97
Open	66.04	66.47	68.21	73.73	81.86	87.52	88.99
Closed	85.96	83.21	81.72	75.48	59.66	34.97	13.21
Shrub	86.12	85.5	84.69	84.48	83.06	83.24	83.33
Non-forest	84.76	84.74	84.08	84.42	83.5	83.39	82.84
<i>Critical</i>							
<i>Per-class producer Kappa (%)</i>							
Woodland	49.12	46.72	46.79	44.78	41.78	38.5	37.46
Open	45.54	47.17	49.36	56.74	68.6	77.34	79.88
Closed	72.42	70.12	67.25	58.27	38.54	14.3	1.66
Shrub	70.53	69.54	69.77	69.16	69.18	68.74	68.56
Non-forest	76.13	75.99	75.89	75.52	75.44	75.43	75.35

balanced case, in the optimal experiment (Table 5). The same ratio also led to optimal per class accuracies in the critical case. Two-class imbalance (between hardest and easiest classes) had only a minor effect on overall multiclass classification accuracy up to 25% (particularly in the critical case), above which accuracy drops steeply with increasing imbalance (Figs. 5 and 6(a)). Mean margins show a similar pattern (though less distinct), whereby certainty in the model (i.e. high margins associated with correctly-classified samples together with low margins associated with misclassified samples), drops steadily above 25% imbalance (Figs. 5 and 6(b)).

Margin-weighted confusion matrices (Table 6) for multiclass model class imbalance, show the degree of certainty associated with correctly classified and misclassified samples. For each class, the highest normalised margin scores are associated with correctly classified instances (i.e. main diagonal cells). Margins for the most difficult classes (open cover, 0.46 and woodland, 0.5) are lower than the closed cover (0.57), shrub (0.75) and non-forest (0.77) classes. A 50% increase and decrease in proportion of open and closed classes improved model classification certainty in the open class by (0.46 to 0.53) and reduced the certainty of the misclassification of open class predicted as closed cover (0.27–0.19). Conversely, the same degree of imbalance reduced certainty in the correct classification of the closed class (0.57–0.43). However, an associated *increase* in the certainty of misclassifying closed as open (between balanced and 50% imbalanced), was less (3%) than the decrease (8%) in certainty of misclassifying open as closed. The latter is closer to a potential shift of the RF ensemble from incorrect to correct majority decisions. In other words, a change in the outcome of misclassified instances.

The results of experiment 2 demonstrate that introducing a *sensible* imbalance can improve ensemble classifier performances.

Unsupervised margin cumulative frequency distribution curves, associated with correctly classified and misclassified instances, comparing balanced and imbalanced training data are shown in Figs. 5 and 6. Curves shifting towards the lower right corner of the plot indicate a higher degree of certainty in correctly classified instances and curves moving towards the upper-left indicate a decrease in the certainty of misclassified instances. Figs. 5 and 6 demonstrates that increasing the degree of imbalance (90% increase worst/decrease best) (Figs. 5 and 6b) results in a higher degree of divergence from the balanced case, relative to 50%

imbalance (Figs. 5 and 6a). The margin distribution pair (correct and misclassified) associated with the evenly balanced case exhibits better behaviour than the distribution pair related to the 50% imbalance, the latter located entirely within the 'leaf-like' pattern of the evenly balanced pair of curves (Figs. 5 and 6a). However, a different pattern is shown in Figs. 5 and 6b comparing balanced and 90% imbalanced cases. The even leaf-shaped pairwise margin distribution only partially contains the imbalanced pairwise margin distribution curves. While the even distribution associated with misclassified instances behaves significantly better than its imbalanced counterpart, the balanced distribution related to correctly classified samples exhibits lower classification confidence than the imbalanced case. This is due to the fact that only 16% of the closed samples have been correctly classified in the extremely imbalanced case (versus 89% for the balanced case). Meanwhile, the accuracy of the open class increased by more than 20% relative to the balanced case. Both imbalanced results have a positive impact on the correctly classified imbalanced margin distribution. On the other hand, the very high misclassification rate of the closed class in the imbalanced case adversely affects the imbalanced misclassified margin distribution.

Comparing pairwise margin distribution curves for the other multiclass imbalance experiment (ratio of open to closed training instances) shows that, in the critical case (Fig. 7b), the divergence of curves from evenly balanced and imbalanced experiments (both correctly classified and misclassified instances) is more pronounced than in the optimal case (Fig. 7a).

6.2. Effect of training data mislabelling on RF performance

6.2.1. Binary mislabelling (experiment 4)

Results of the binary classification class mislabelling experiment are shown in Table 7 and Fig. 8. For the optimal case, the impact of the introduction of mislabeled data (i.e. forest class mislabeled as non-forest and non-forest as forest) on overall classification accuracy was negligible (only a 2.8% reduction from 0% mislabeled to 25% mislabeled). For the smaller (critical) training size, the reduction in overall accuracy was higher (a 6.9% accuracy reduction from 0% mislabeled to 25% mislabeled). The classifier uncertainty statistic (mean margin) showed a greater reduction associated with an increasing number of mislabeled training instances. Mean margin decreases with each incremental increase in the proportion of mislabeled training instances for both optimal and critical cases (Fig. 8). Results showing this decline in overall model certainty is supported by a reduction in the margin mode (Table 7). For the optimal case, with each increase in the proportion of mislabeled instances, the dominant margin reduced from 0.86 to 0.30 for 25% mislabeled instances. Loss in classification confidence is more pronounced for the critical case.

6.2.2. Multiclass mislabelling (experiment 5)

Between *clean* training data and 25% class mislabelling, overall classification accuracy was reduced by 6.6% for the optimal case and 7.2% for the critical case (Table 8). However, associated with this was about a 55% decrease in the mean margin for both optimal and critical cases, which indicates that while training data *noise* has only a minor impact on accuracy relative to the amount of mislabeled training data, it does have a strong influence on classification uncertainty. Dominant margin results for the optimal case support this – from 10% to 25% mislabelling, the dominant margin drops from 0.52 to 0.18 (Table 8).

Fig. 9 shows change in classification accuracy (a) and mean margin (b) with mislabeled training instances. By selecting a random subset of training instances to build each decision tree in the ensemble, as well as randomly selecting the features involved in data partitioning at each tree node, RF is robust to noise. This is

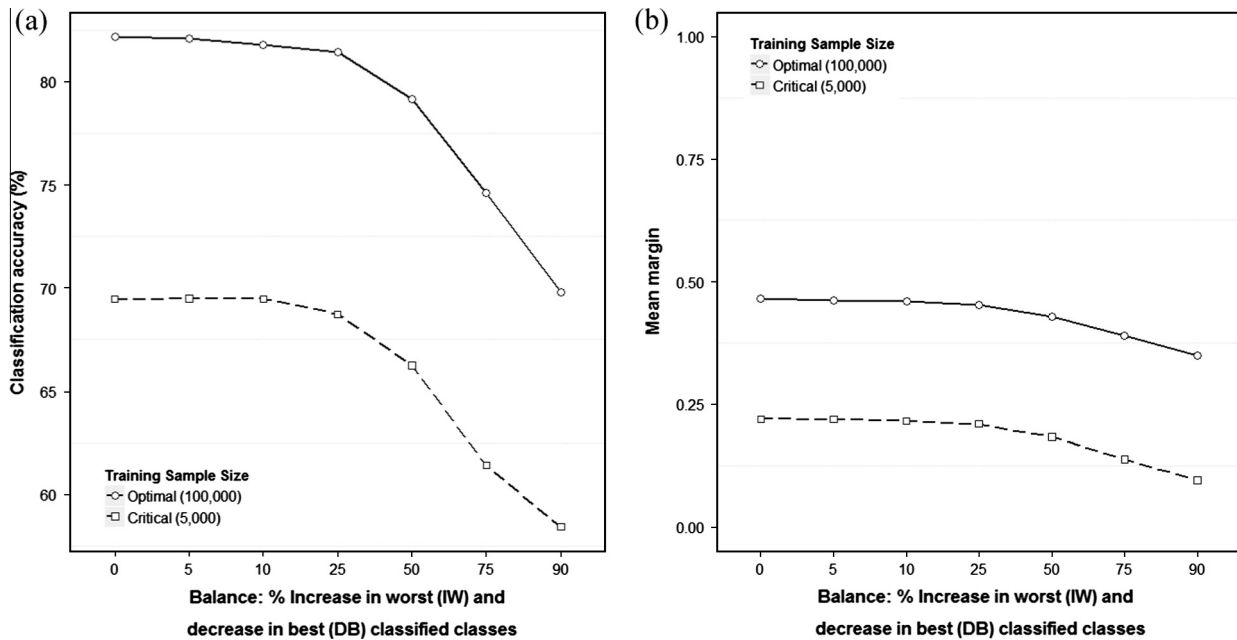


Fig. 5. Effect of multiclass imbalance on overall multiclass classification accuracy (a) and mean margin (b).

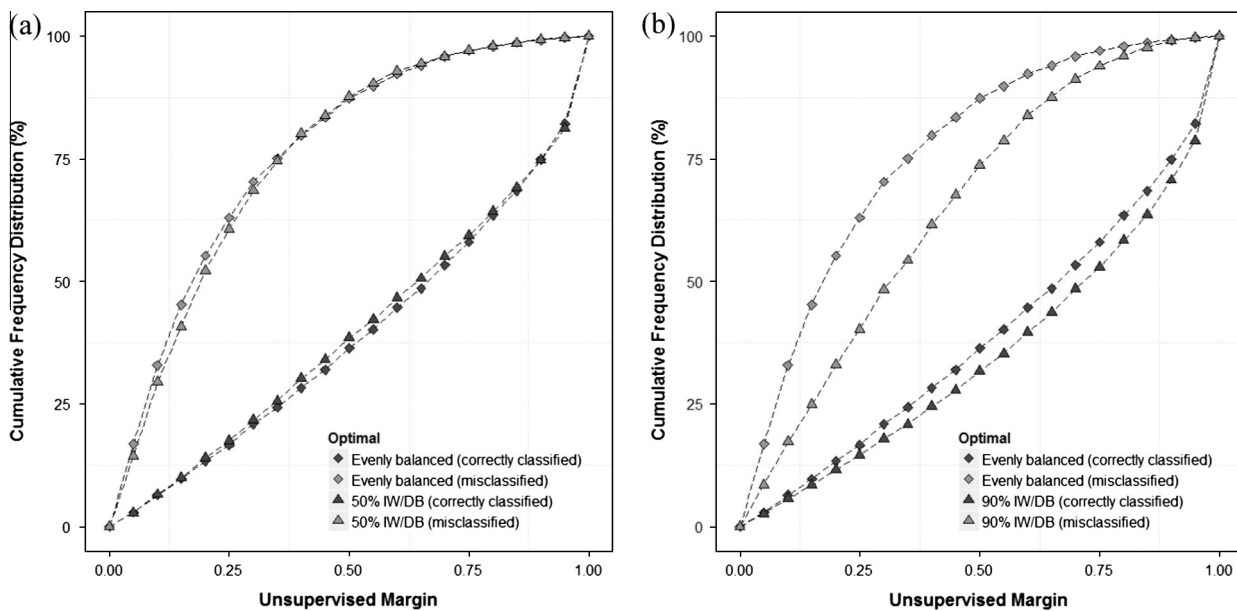


Fig. 6. Unsupervised margin cumulative frequency distribution curves associated with correctly and misclassified instances, comparing balanced versus 50% increase/decrease open/closed (a) and balanced versus 90% increase/decrease open/closed (b).

further demonstrated by comparison to the accuracy of a classifier comprising only a single decision tree constructed using all available training instances (optimal case) and not a bootstrapped sample (reduced by about a third in standard bagging) (Fig. 9a). Results of the single tree classifier show a lower overall accuracy (17% less than the optimal ensemble case) and a steeper reduction in overall accuracy associated with the increasing proportion of mislabeled training instances. Despite a dramatic reduction in training data, the accuracy curve of RF for the critical case behaves significantly better than the single curve. This again, highlights the capability of ensemble classifiers, especially RF, in handling multiple mislabelling classification problems.

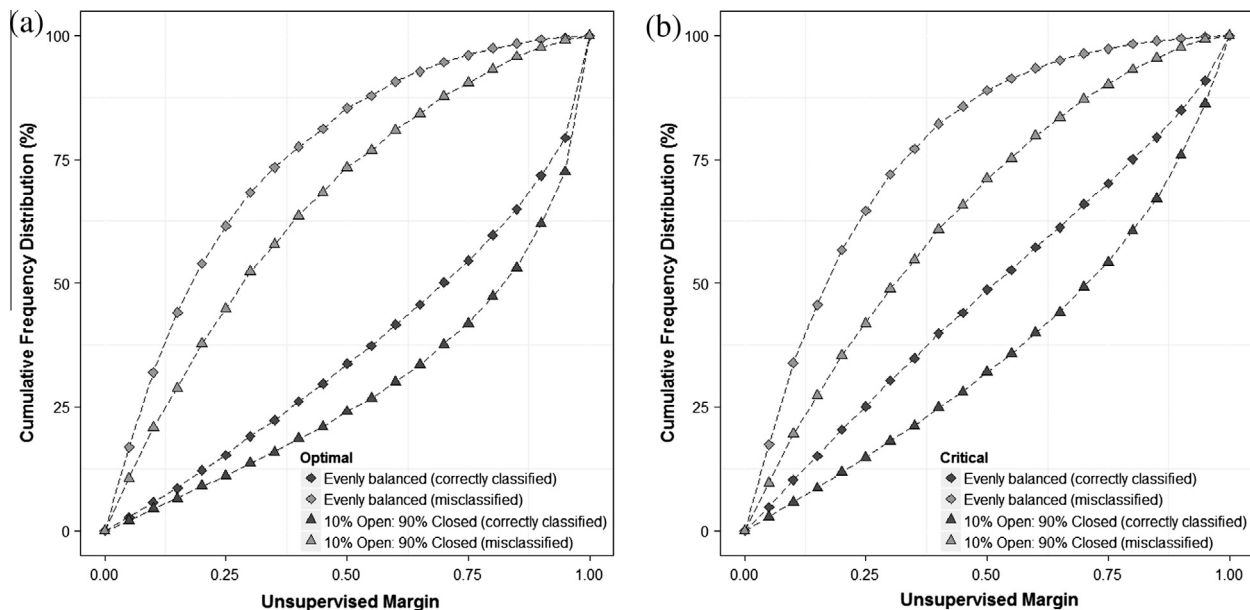
Margin weighted confusion matrices (optimal case, Table 9, and critical case, Table 10) demonstrate the increase in per-class

uncertainty (lower classification confidence values) associated with correctly classified instances (main diagonals) with an increasing proportion of mislabeled instances. For the critical case in particular, where a training set is small, the addition of even low levels of class mislabelling can greatly increase ensemble diversity and the likelihood of more randomness in the classifier outcome. At the 25% mislabelling level, both optimal and critical cases have similar margins. Margin entropy values (Table 8) are initially (0% artificial mislabelling) very high (close to maximum entropy), reflecting a high diversity in the large area forest data. The margin entropy results should be interpreted with caution. In the optimal case, the margin entropy increases from the original (0% mislabelling) to 5% mislabelling. An increase in entropy with noise is an expected behaviour, as uncertainty increases with noise. In

Table 6

Multiclass imbalance confusion matrices and margin-weighted confusion matrices for optimal case (balanced and 50% imbalanced).

	Confusion Matrix					Margin-weighted confusion matrix				
	Observed (test) data									
	Evenly balanced training samples									
	Woodland	Open	Closed	Shrub	Non-forest	Woodland	Open	Closed	Shrub	Non-forest
Prediction										
Woodland	3714	449	121	261	265	0.50	0.20	0.17	0.21	0.28
Open	512	3559	335	164	129	0.18	0.46	0.20	0.19	0.22
Closed	245	784	4434	110	63	0.18	0.27	0.57	0.18	0.21
Shrub	236	73	49	4379	168	0.32	0.16	0.13	0.75	0.40
Non-forest	293	135	61	86	4375	0.32	0.34	0.28	0.23	0.77
50% increase in Open canopy samples and 50% reduction in Closed canopy samples										
	Woodland	Open	Closed	Shrub	Non-forest	Woodland	Open	Closed	Shrub	Non-forest
Prediction										
Woodland	3478	234	139	230	237	0.47	0.22	0.19	0.20	0.27
Open	937	4370	1430	291	226	0.23	0.53	0.23	0.23	0.24
Closed	61	220	3282	27	21	0.18	0.19	0.43	0.15	0.20
Shrub	225	58	64	4369	179	0.33	0.14	0.18	0.75	0.37
Non-forest	299	118	85	83	4337	0.30	0.35	0.29	0.24	0.77

**Fig. 7.** Unsupervised margin cumulative frequency distribution curves, comparing balanced and ratio-imbalanced (10 open: 90 closed) for optimal (a) and critical (b) cases.**Table 7**

RF model performance results for optimal and critical binary class mislabelling experiments.

% of Misabeled training instances (per class)							
	0%	2.5%	5%	10%	15%	20%	25%
Overall Kappa (%)							
Optimal	81.04	80.96	81.12	80.08	79.12	78.32	75.36
Critical	68.42	67.34	67.26	64.59	63.13	56.65	54.64
Margin entropy							
Optimal	2.67	2.91	3.07	3.11	3.11	3.04	2.90
Critical	3.18	3.23	3.26	3.28	3.26	3.10	2.93
Dominant margin							
Optimal	0.86	0.85	0.77	0.67	0.58	0.48	0.30
Critical	0.79	0.70	0.70	0.61	0.52	0.34	0.18
Per-class producer Kappa (%)							
Optimal							
Forest	82.96	83.09	83.11	82.66	80.87	80.05	78.37
Non-forest	79.2	78.94	79.22	77.66	77.45	76.66	72.57
Critical							
Forest	70.41	69.22	68.79	66.22	66.94	56.45	56.9
Non-forest	66.53	65.57	65.79	63.04	59.73	56.84	52.55

the meantime, the mean margin decreases (Fig. 9b), suggesting an increase in lower margins to achieve a more balanced distribution between higher and lower margins, and consequently an increase in entropy. Then, with increasing mislabeled instances (i.e. more noise), entropy decreases as lower margins become over-represented (as illustrated by the decreasing mean margin) inducing an increasing imbalance in margin frequencies.

Results of the binary (experiment 4) and multi-class mislabelling (experiment 5) experiments support previous research demonstrating RF resistance to mislabelling (Rodriguez-Galiano et al., 2012; Rogan et al., 2008). While results of these experiments are encouraging for noise resistance, ensemble margin statistics (mean and dominant margins) and MWCM reveal mislabelling to have a large effect on classification uncertainty.

Results on the affect of training data characteristics on random forest performance inform the design and implementation of large area land cover classification for natural resource management. The study's findings highlight issues for consideration in the design, training data collection and RF model construction phases

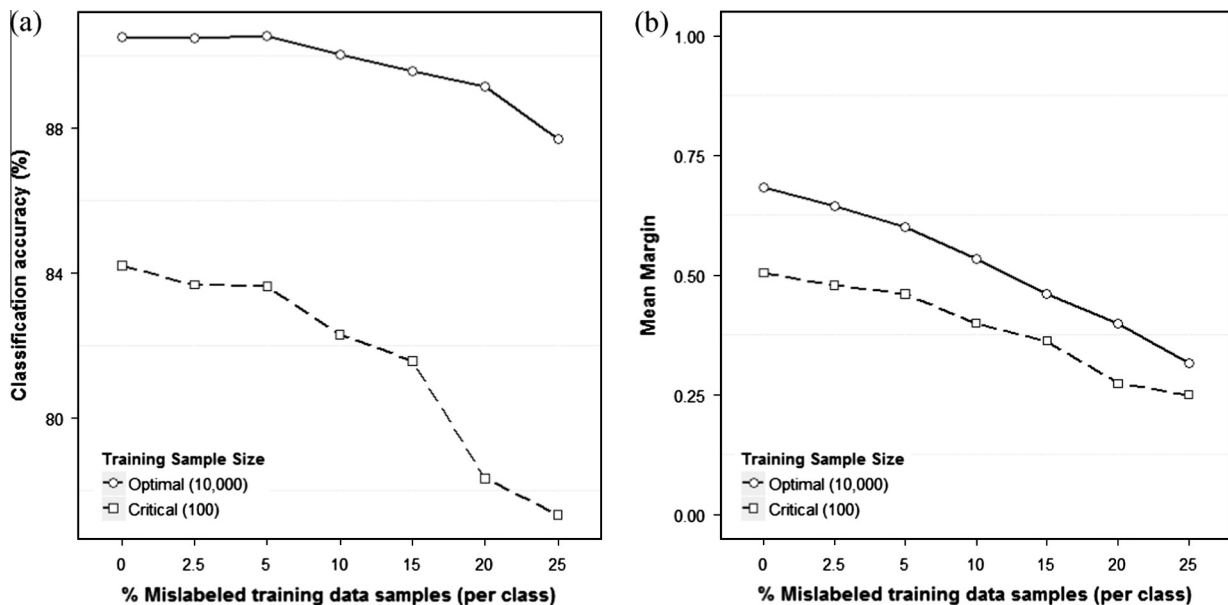


Fig. 8. Effect of class mislabelling on binary classification overall accuracy (a) and mean margin (b).

Table 8

RF model performance results for optimal and critical size multiclass class mislabelling experiments.

	% of Mislabeled training instances in each class						
	0%	2.5%	5%	10%	15%	20%	25%
Overall Kappa (%)							
Optimal	78.3	77.1	76.8	76	74.4	72.7	70.1
Critical	62.61	61.7	61	59.9	58.4	56.4	53.6
Margin entropy							
Optimal	3.24	3.3	3.31	3.25	3.12	2.96	2.79
Critical	3.23	3.22	3.18	3.09	2.95	2.79	2.61
Dominant margin							
Optimal	0.78	0.77	0.68	0.52	0.43	0.26	0.18
Critical	0.65	0.02	0.02	0.02	0.02	0.02	0.01

of a classification. Furthermore, these are issues which inform where best to allocate limited resources in building a robust and accurate RF classification. While results demonstrate that balanced distributions of training data achieve the greatest overall classification accuracy and certainty, introducing imbalance favouring more difficult to classify classes, can be used to boost per class accuracy without compromising a classification's overall accuracy. Results from mislabelling experiments emphasize the importance of training data labelling accuracy in large area classifications. While the accuracy of random forests is relatively robust to noise in training data, associated model certainty is less so. As such, assuming that increasing that amount of training data collected leads to an increase in the proportion of mislabelled training data instances, it is important to consider an appropriate balance between the amount of training data (per class), class mislabelling, classification accuracy and uncertainty.

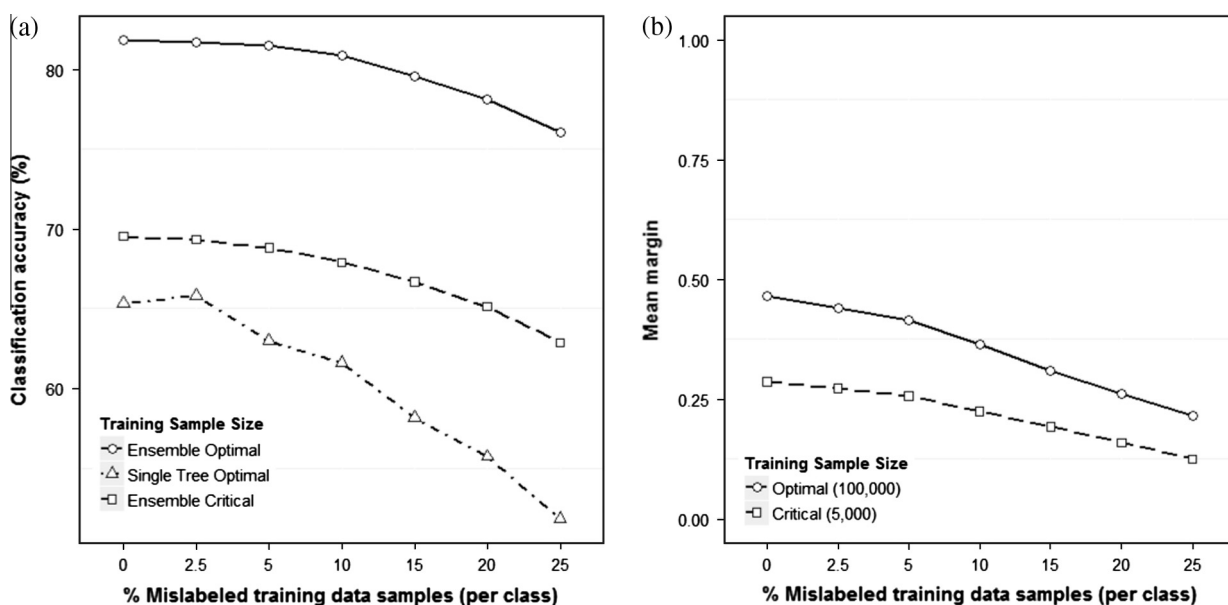


Fig. 9. Effect of class mislabelling on multiclass classification overall accuracy (a) and mean margin (b).

Table 9

Multiclass mislabelling confusion matrices and margin weighted confusion matrices for optimal case.

	Confusion matrix					Margin-weighted confusion matrix				
	Observed (test) data									
	5% mislabeled training instances per class									
	Woodland	Open	Closed	Shrub	Non-forest	Woodland	Open	Closed	Shrub	Non-forest
<i>Prediction</i>										
Woodland	3658	439	104	287	300	0.45	0.18	0.14	0.21	0.26
Open	619	3590	414	173	139	0.17	0.42	0.17	0.17	0.19
Closed	253	763	4370	108	62	0.19	0.25	0.50	0.15	0.20
Shrub	202	70	45	4355	168	0.31	0.13	0.13	0.67	0.37
Non-forest	268	138	67	77	4331	0.30	0.32	0.26	0.23	0.69
	25% mislabeled training instances per class									
	Woodland	Open	Closed	Shrub	Non-forest	Woodland	Open	Closed	Shrub	Non-forest
<i>Prediction</i>										
Woodland	3352	359	90	658	634	0.29	0.13	0.11	0.15	0.17
Open	1088	3677	782	251	154	0.15	0.26	0.12	0.12	0.12
Closed	240	825	4051	104	63	0.12	0.14	0.28	0.10	0.11
Shrub	138	35	22	3935	146	0.19	0.10	0.10	0.38	0.23
Non-forest	182	104	55	52	4003	0.19	0.19	0.19	0.17	0.39

Table 10

Multiclass mislabelling confusion matrices and margin weighted confusion matrices for critical case.

	Confusion matrix					Margin-weighted confusion matrix				
	Observed (test) data									
	5% mislabeled training instances per class									
	Woodland	Open	Closed	Shrub	Non-forest	Woodland	Open	Closed	Shrub	Non-forest
<i>Prediction</i>										
Woodland	2850	693	180	552	461	0.31	0.18	0.12	0.19	0.21
Open	979	2872	813	338	169	0.15	0.27	0.15	0.13	0.14
Closed	505	1219	3869	294	115	0.19	0.25	0.40	0.18	0.15
Shrub	285	67	58	3662	305	0.30	0.10	0.11	0.63	0.37
Non-forest	381	150	79	154	3950	0.28	0.30	0.25	0.21	0.61
	25% mislabeled training instances per class									
	Woodland	Open	Closed	Shrub	Non-forest	Woodland	Open	Closed	Shrub	Non-forest
<i>Prediction</i>										
Woodland	2576	602	158	921	971	0.24	0.15	0.10	0.16	0.16
Open	1540	3092	1354	503	242	0.14	0.21	0.13	0.12	0.11
Closed	446	1179	3409	271	102	0.13	0.16	0.24	0.12	0.11
Shrub	194	30	28	3208	254	0.20	0.09	0.07	0.36	0.24
Non-forest	243	98	50	96	3432	0.18	0.20	0.17	0.15	0.36

7. Conclusion

Results from this study provide important insights into the behaviour of the RF ensemble classifier that should provide a guide to the design of an operational implementation in other large area settings, particularly across complex, dynamic and heterogeneous environments. Measures of accuracy and confidence reveal the degree of influence that training data imbalance and mislabelling have on overall and per-class classification performance. The binary and multiclass land cover classification experiments showed the relevance of the introduced ensemble margin criteria and margin weighted confusion matrix for the investigation of both imbalance and mislabelling problems in ensemble classification.

Across large areas with spectrally similar and noisy land cover classes, a degree of training class mislabelling is inevitable. Our findings reveal that while traditional confusion matrices (derived either from independent validation data or an Out-Of-Bag (OOB) sample) can show reasonable classification performance, classification certainty can be significantly reduced, especially where the amount of training data is limited. While previous studies have shown classification to perform better with balanced datasets (Estabrooks et al., 2004; Freeman et al., 2012), we demonstrate that deliberately imbalancing classes can be used to improve the

classification and performance of more challenging classes, without significantly compromising overall and other per-class classification results. Given the costs of training data collection (ground-based collection or from high resolution remote sensing data), in an operational setting, optimising a classification involves balancing the total amount, class distribution and labelling accuracy of training data. For example, prioritising rare or more difficult classes over those that are more common or easy to classify. And, where resources limit the amount of available training data, the quality (i.e. correct labelling) becomes a more important consideration.

Future research will investigate methods to address imbalance and mislabelling problems and using margin statistics and the MWCM to evaluate their success.

References

- Boyd, D.S., Danson, F.M., 2005. Satellite remote sensing of forest resources: three decades of research development. *Prog. Phys. Geogr.* 29, 1–26. http://dx.doi.org/10.1191/0309133305_pp432ra.
- Bradley, C., Friedl, M., 1996. Identifying and Eliminating Mislabeled Training Instances. In: *Proceedings of the Thirteenth National Conference on Artificial Intelligence*, Amherst, MA, pp. 799–805.
- Breiman, L., 1996. Bagging predictors. *Mach. Learn.* 24, 123–140.

- Breiman, L., 2001. Random forests. *Mach. Learn.* 45, 5–32. <http://dx.doi.org/10.1023/A:1010933404324>.
- Carletta, J., 1996. Assessing agreement on classification tasks: the Kappa statistic. *Comput. Linguist.* 22, 249–254.
- Chawla, N.V., Bowyer, K.W., Hall, L.O., Kegelmeyer, W.P., 2002. SMOTE: Synthetic Minority Over-sampling Technique. *J. Artif. Intell. Res.* 16, 321–357.
- Chen, C., Liaw, A., Breiman, L., 2004. Using Random Forest to Learn Imbalanced Data. Statistics Department. University of California, Berkeley.
- CSIRO, 2011. One-second SRTM digital elevation model. <<http://www.csiro.au/Outcomes/Water/Water-information-systems/One-second-SRTM-Digital-Elevation-Model.aspx>>.
- Cutler, D., Jr., T.E., Beard, K., 2007. Random forests for classification in ecology. *Ecology* 88, 2783–2792.
- DeFries, R., Cheung-Wai Chan, J., 2000. Multiple criteria for evaluating machine learning algorithms for land cover classification from satellite data. *Remote Sens. Environ.* 74, 503–515. [http://dx.doi.org/10.1016/S0034-4257\(00\)00142-5](http://dx.doi.org/10.1016/S0034-4257(00)00142-5).
- Delaney, J.L., Skidmore, A.K., 1998. Discrepancy or error in forest type classification. *Aust. For.* 61, 82–88. <http://dx.doi.org/10.1080/00049158.1998.10674724>.
- Department of Agriculture Fisheries and Forestry, 2012. Australia's forest at a glance. Canberra.
- Estabrooks, A., Jo, T., Japkowicz, N., 2004. A multiple resampling method for learning from imbalanced data sets. *Comput. Intell.* 20, 18–36. <http://dx.doi.org/10.1111/j.0824-7935.2004.t01-1-00228.x>.
- Farmer, E., Jones, S., Clarke, C., Buxton, L., Soto-Berelov, M., Page, S., Mellor, A., Haywood, A., 2013. Creating a large area landcover dataset for public land monitoring and reporting. In: Arrowsmith, C., Bellman, C., Cartwright, W., Jones, S., Shortis, M. (Eds.), *Progress in Geospatial Science Research*. Publishing Solutions, Melbourne, pp. 85–98.
- Flood, N., Danaher, T., Gill, T., Gillingham, S., 2013. An operational scheme for deriving standardised surface reflectance from landsat TM/ETM+ and SPOT HRG Imagery for Eastern Australia. *Remote Sens.* 5, 83–109. <http://dx.doi.org/10.3390/rs5010083>.
- Food and Agriculture Organization of the United Nations, 2001. Global forest resources assessment, 2000.
- Foody, G., 1999. The significance of border training patterns in classification by a feedforward neural network using back propagation learning. *Int. J. Remote Sens.* 20, 37–41.
- Foody, G.M., 2002. Status of land cover classification accuracy assessment. *Remote Sens. Environ.* 80, 185–201.
- Foody, G., 2004. Thematic map comparison: evaluating the statistical significance of differences in classification accuracy. *Photogramm. Eng. Remote Sensing* 70, 627–633.
- Franklin, J., 1995. Predictive vegetation mapping: geographic modelling of biospatial patterns in relation to environmental gradients. *Prog. Phys. Geogr.* 19, 474–499.
- Freeman, E.A., Moisen, G.G., Frescino, T.S., 2012. Evaluating effectiveness of down-sampling for stratified designs and unbalanced prevalence in Random Forest models of tree species distributions in Nevada. *Ecol. Modell.* 233, 1–10. <http://dx.doi.org/10.1016/j.ecolmodel.2012.03.007>.
- Gislason, P., Benediktsson, J., Sveinsson, J., 2006. Random Forests for land cover classification. *Pattern Recognit. Lett.* 27, 294–300. <http://dx.doi.org/10.1016/j.patrec.2005.08.011>.
- Guisan, A., Zimmermann, N.E., 2000. Predictive habitat distribution models in ecology. *Ecol. Modell.* 135, 147–186. [http://dx.doi.org/10.1016/S0304-3800\(00\)00354-9](http://dx.doi.org/10.1016/S0304-3800(00)00354-9).
- Guo, L., 2011. Margin framework for ensemble classifiers. Application to remote sensing data. University of Bordeaux.
- Guo, L., Chehata, N., Mallet, C., Boukir, S., 2011. Relevance of airborne lidar and multispectral image data for urban scene classification using Random Forests. *ISPRS J. Photogramm. Remote Sens.* 66, 56–66. <http://dx.doi.org/10.1016/j.isprsjprs.2010.08.007>.
- Haralick, R.M., 1979. Statistical and structural approach to texture. *Proc. IEEE* 67, 786–804.
- He, H., Garcia, E.A., 2009. Learning from imbalanced data. *IEEE Trans. Knowl. Data Eng.* 21, 1263–1284. <http://dx.doi.org/10.1109/TKDE.2008.239>.
- Houlder, D., 2001. ANUCLIM (version 5.1).
- Japkowicz, N., Stephen, S., 2002. The class imbalance problem: a systematic study. *Intell. Data Anal.* 6, 429–450.
- Kayitakire, F., Hamel, C., Defourny, P., 2006. Retrieving forest structure variables based on image texture analysis and IKONOS-2 imagery. *Remote Sens. Environ.* 102, 390–401. <http://dx.doi.org/10.1016/j.rse.2006.02.022>.
- Kuncheva, L., Whitaker, C., 2003. Measures of diversity in classifier ensembles and their relationship with the ensemble accuracy. *Mach. Learn.* 51, 181–207.
- Liaw, A., Wiener, M., 2002. Classification and Regression by randomForest. *R News* 2, 18–22.
- Lillesand, T.M., Kiefer, R.W., 1994. *Remote Sensing and Image Interpretation*, third ed. John Wiley & Sons Ltd.
- Ling, C., Li, C., 1998. Data mining for direct marketing problems and solutions. In: *Proceedings of the Fourth International Conference on Knowledge Discovery and Data Mining*. New York.
- Lippitt, C., Rogan, J., Li, Z., 2008. Mapping selective logging in mixed deciduous forest: a comparison of machine learning algorithms. *Photogramm. Eng. Remote Sens.* 74, 1201–1211.
- Lowell, K., Woodgate, P., Richards, G., Jones, S.D., Buxton, L., 2005. Fuzzy reliability assessment of multi-period land-cover change maps. *Photogramm. Eng. Remote Sens.* 71, 939–945.
- McRoberts, R.E., 2011. Satellite image-based maps: scientific inference or pretty pictures? *Remote Sens. Environ.* 115, 715–724. <http://dx.doi.org/10.1016/j.rse.2010.10.013>.
- Mellor, A., Haywood, A., 2010. Remote sensing Victoria's public land forests – a two tiered synoptic approach. In: *Proceedings of the 15th Australian Remote Sensing and Photogrammetry Conference*. Alice Springs.
- Mellor, A., Haywood, A., Stone, C., Jones, S., 2013. The performance of random forests in an operational setting for large area sclerophyll forest classification. *Remote Sens.* 5, 2838–2856. <http://dx.doi.org/10.3390/rs5062838>.
- Morgan, J.L., Gergel, S.E., Coops, N.C., 2010. Aerial photography: a rapidly evolving tool for ecological management. *Bioscience* 60, 47–59. <http://dx.doi.org/10.1525/bio.2010.60.1.9>.
- Na, X., Zang, S., Wang, J., 2009. Evaluation of Random Forest Ensemble Classification for Land Cover Mapping Using TM and Ancillary Geographical Data. 2009 Sixth Int. Conf. Fuzzy Syst. Knowl. Discov. pp. 89–93. <http://dx.doi.org/10.1109/FSKD.2009.165>.
- Beaumont, L., Hughes, L., Poulsen, M., 2005. Predicting species distributions: use of climatic parameters in BIOCLIM and its impact on predictions of species' current and future distributions. *Ecol. Modell.* 186, 250–269.
- Paget, M.J., King, E.A., 2008. MODIS land data sets for the Australian region. Canberra.
- Pal, M., 2005. Random forest classifier for remote sensing classification. *Int. J. Remote Sens.* 26, 217–222. <http://dx.doi.org/10.1080/01431160412331269698>.
- Provost, F., 2000. Machine Learning from Imbalanced Data Sets 101 (Extended Abstract). In: *Proceedings of the AAAI-2000 Workshop on Imbalanced Data Sets*.
- R Development Core Team, 2011. R: A Language and Environment for Statistical Computing.
- Rodríguez-Galiano, V.F., Abarca-Hernández, F., Ghimire, B., Chica-Olmo, M., Atkinson, P.M., Jeganathan, C., 2011. Incorporating spatial variability measures in land-cover classification using random forest. *Procedia Environ. Sci.* 3, 44–49.
- Rodríguez-Galiano, V.F., Ghimire, B., Rogan, J., Chica-Olmo, M., Rigol-Sanchez, J.P., 2012. An assessment of the effectiveness of a random forest classifier for land-cover classification. *ISPRS J. Photogramm. Remote Sens.* 67, 93–104. <http://dx.doi.org/10.1016/j.isprsjprs.2011.11.002>.
- Rogan, J., Franklin, J., Stow, D., Miller, J., Woodcock, C., Roberts, D., 2008. Mapping land-cover modifications over large areas: a comparison of machine learning algorithms. *Remote Sens. Environ.* 112, 2272–2283. <http://dx.doi.org/10.1016/j.rse.2007.10.004>.
- Schapire, R.E., Freund, Y., Bartlett, P., Lee, W.S., 1998. Boosting the margin: a new explanation for the effectiveness of voting methods. *Ann. Stat.* 26, 1651–1686.
- Shannon, C.E., 1948. A mathematical theory of communication. *Bell Syst. Tech. J.* 27, 379–423.
- Sluban, B., Gamberger, D., Lavrač, N., 2014. Ensemble-based noise detection: noise ranking and visual performance evaluation. *Data Min. Knowl. Discov.* 28, 265–303.
- Tsoumakas, G., Partalas, I., Vlahavas, I., 2009. Applications of supervised and unsupervised ensemble methods. In: *An Ensemble Pruning Primer*. Springer, pp. 1–13.
- Weiss, G.M., Provost, F., 2003. Learning when training data are costly: the effect of class distribution on tree induction. *J. Artif. Intell. Res.* 19, 315–354.
- Wulder, M., 1998. Optical remote-sensing techniques for the assessment of forest inventory and biophysical parameters. *Prog. Phys. Geogr.* 22, 449.